Trend Rotation and Multi-Asset Momentum

Two systematic strategies compared: SPY/IEF moving-average rotation and a six-ETF monthly momentum rotation

Prepared October 19, 2026

Hypothetical backtested results. For discussion purposes only. Generated by an AI multi-agent research pipeline.

Executive Summary

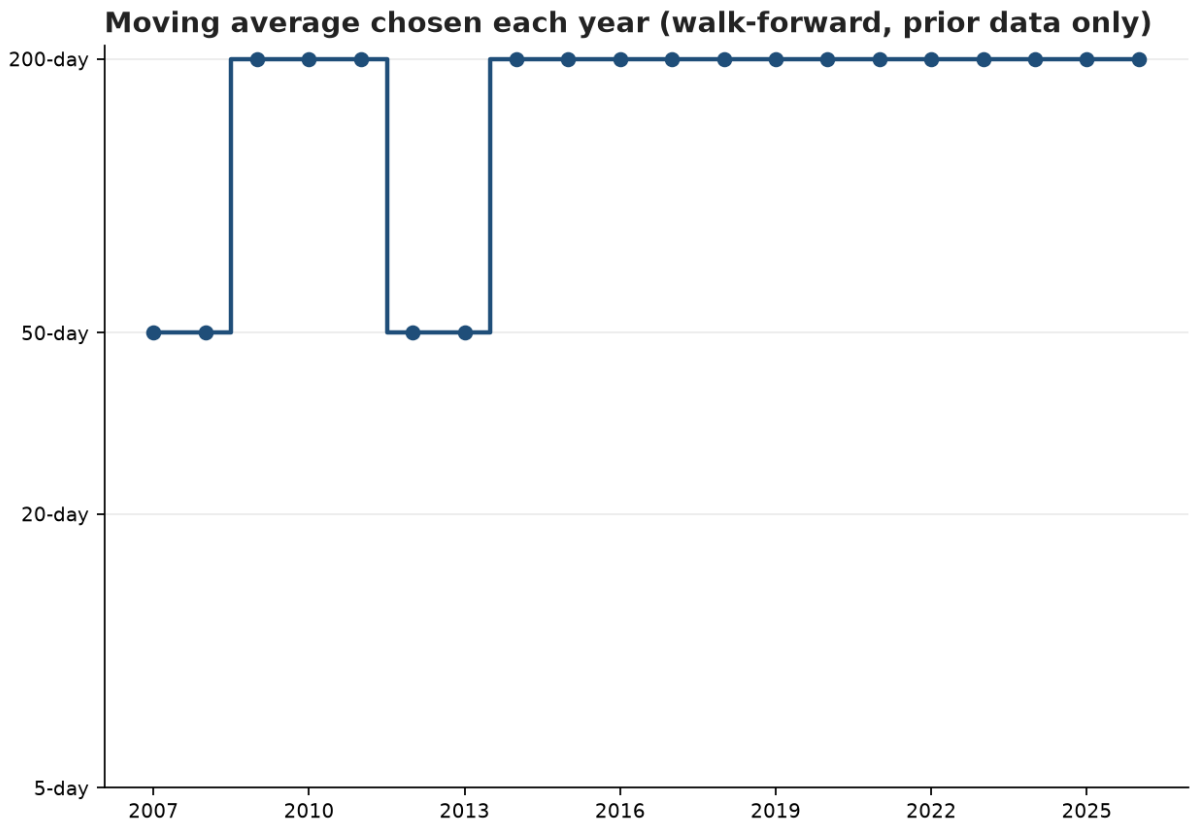
- Return verdict since 2011: SPY (14.1% CAGR) and QQQ (18.7%) beat every momentum lookback (9.2% to 11.9%, before costs) and the MA strategy (9.3%, net of costs).
- Sharpe since 2011: momentum 0.58 to 0.77 and MA strategy 0.67 vs SPY 0.77 and QQQ 0.86. The best strategy Sharpe (3-Month momentum) only matched SPY; QQQ was highest.
- Drawdown is where the strategies differed: max drawdown -24.1% (3-Month) to -32.9% (1-Month) and MA -26.4%, vs SPY -33.7% and QQQ -35.1%. 3-Month was shallowest.
- MA strategy since 2007: max drawdown -26.4% vs SPY -55.2%, but CAGR 7.6% vs 10.9% and Sharpe 0.53 vs 0.55. A 60/40 mix beat it on CAGR (8.4%) and Sharpe (0.63).
- Recent windows also favored SPY: over 5Y the MA strategy had a deeper drawdown (-26.4% vs -24.5%) and lower Sharpe (0.39 vs 0.57); over 1Y its Sharpe was 0.78 vs 1.00.
- Momentum is always fully invested in three ETFs (50/30/20) with no defensive exit and excludes trading costs; at MA-level costs its CAGR would be 9.1% to 11.7%.

The Problem: Deep Equity Drawdowns

- SPY's max drawdown since 2007 was -55.2%, and its worst rolling 1-year return was -47.4%.
- Calendar 2008: SPY returned -36.8%. In 2022, SPY returned -18.2% and a 60/40 mix also fell -16.4%.
- Large losses test investor discipline; selling during a drawdown can lock in losses.
- The MA strategy addresses this with a mechanical, rules-based exit from SPY into intermediate Treasuries (IEF).
- The trade-off: a tactical SPY/Treasury allocation captures only part of SPY's upside, as the following slides show.

How the MA Strategy Works: Rules and Walk-Forward Selection

- Holds SPY when SPY is above its moving average; holds IEF (intermediate Treasuries) otherwise.
- Candidate averages: 5, 20, 50 and 200 days. The average is re-selected each January using only prior data (walk-forward).
- Over 20 years of walk-forward selection, the 200-day was chosen 16 times and the 50-day 4 times.
- 158 switches since 2007, with 77.0% of time in SPY. Results are net of modeled trading costs.
- Current allocation: 100.0% SPY.



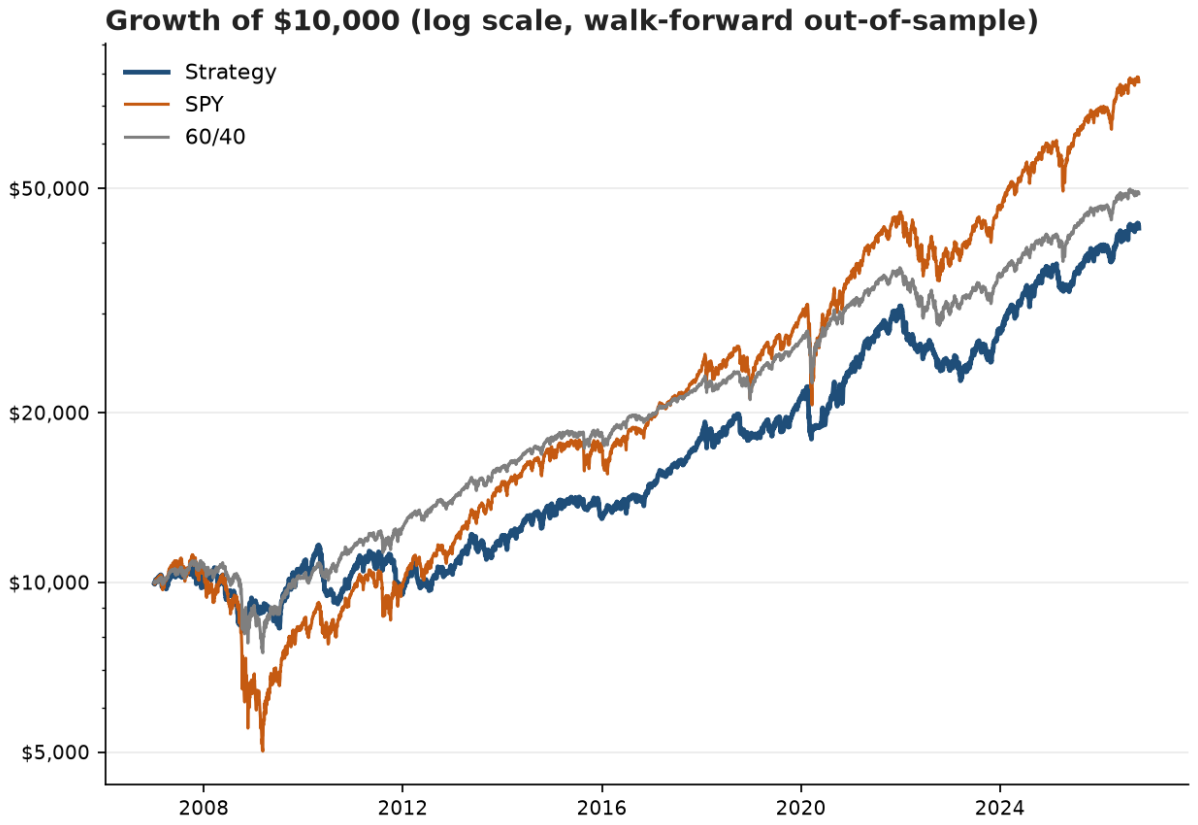
Context: Each Moving Average Over the Full Period (IN-SAMPLE)

Rule (IN-SAMPLE)	CAGR	Volatility	Sharpe	Max DD	Switches	% in SPY	Years chosen
5-day MA switch	-3.5%	12.9%	-0.33	-66.5%	1330	59.4%	0
20-day MA switch	4.0%	12.3%	0.26	-37.1%	570	66.3%	0
50-day MA switch	6.0%	12.0%	0.42	-29.8%	348	70.2%	4
200-day MA switch	9.4%	12.5%	0.66	-26.4%	112	79.1%	16

IN-SAMPLE (2007-01-03 to 2026-10-16): these rows use hindsight. Picking the best row after the fact would overstate results. The strategy instead picks a moving average each January using only earlier data, and that walk-forward result is what the performance pages show. "Years chosen" counts how many years the walk-forward selected each rule.

MA Strategy vs SPY Since 2007: Growth of $10,000

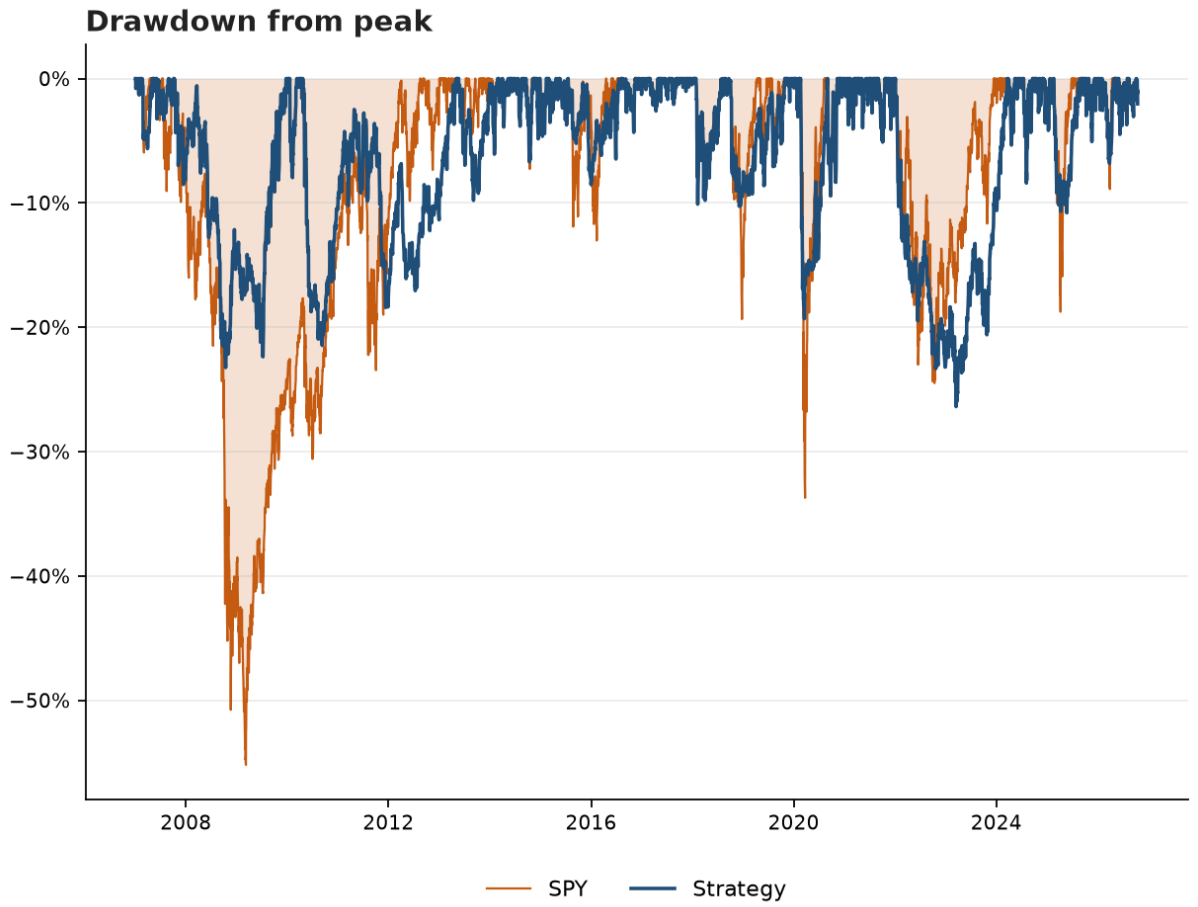
- Growth of $10,000 since 2007: $42,495 for the MA strategy vs $77,253 for SPY and $48,905 for 60/40.
- CAGR 7.6% vs SPY 10.9%, with volatility of 12.4% vs 19.5%.
- Up capture 54.9% and down capture 53.5%: it participated in roughly half of SPY's moves in both directions.
- Sharpe 0.53 vs SPY 0.55: lower volatility did not translate into a better risk-adjusted return over the full period.
- Calmar 0.29 vs SPY 0.20, reflecting the much smaller max drawdown.



Hypothetical backtested performance. Not actual trading; not net of fees. MA strategy results are net of modeled trading costs; momentum results exclude trading costs. Past performance does not guarantee future results. See Important Disclosures.

MA Strategy Risk: Drawdowns vs SPY Since 2007

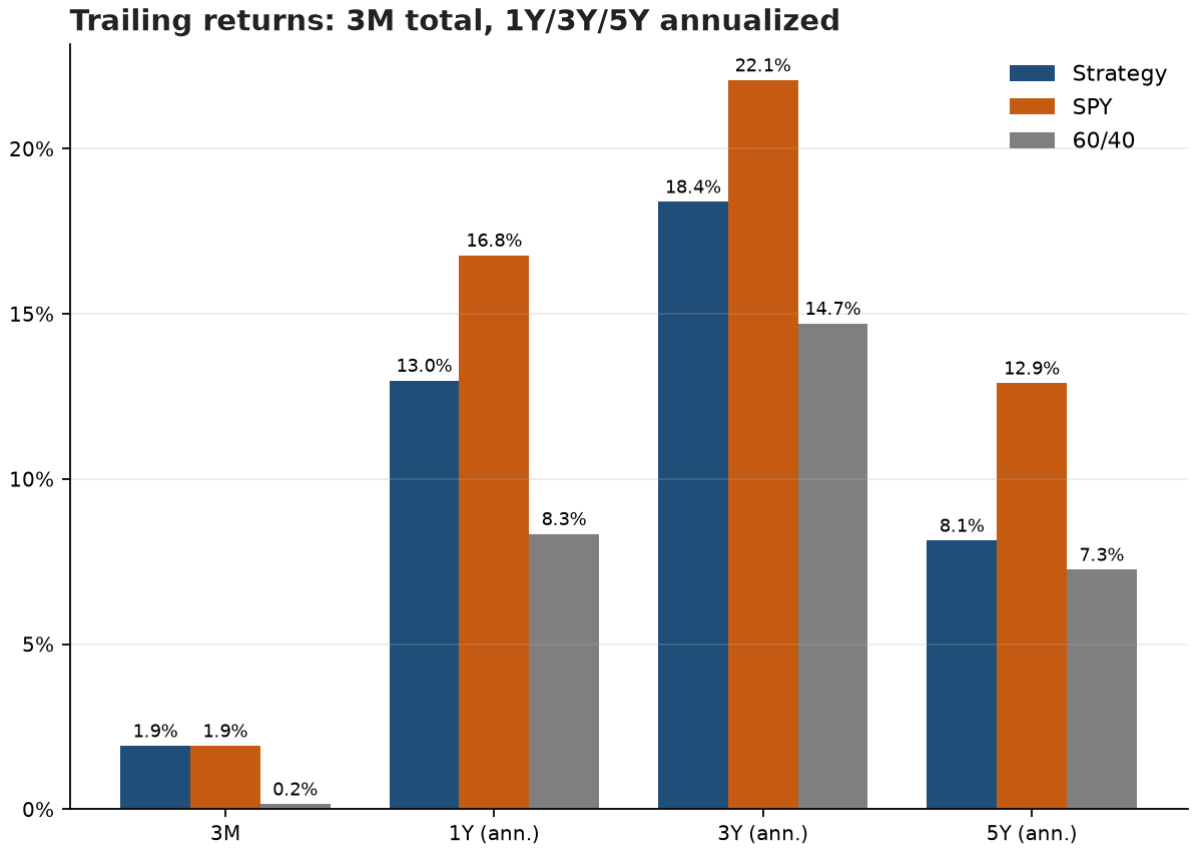
- Max drawdown -26.4% vs SPY -55.2% and 60/40 -31.4% since 2007.
- Worst rolling 1-year return -23.1% vs SPY -47.4%; best 44.6% vs SPY 77.5%.
- 2008: -6.8% vs SPY -36.8%. 2018: 1.6% vs SPY -4.6%.
- Protection was not consistent: in 2022 it lost -22.7% vs SPY -18.2%, as the Treasury exit offered little help when bonds also fell.
- Over 5Y its max drawdown was deeper than SPY's (-26.4% vs -24.5%). Worst day since 2007: -5.8% vs SPY -10.9%.



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MA Strategy Performance by Period: 3M, 1Y, 3Y, 5Y

- 3M: 1.9% vs SPY 1.9%. The strategy was fully in SPY with no switches, so results were identical.
- 1Y: 13.0% vs SPY 16.8%; Sharpe 0.78 vs 1.00; max drawdown -6.8% vs -8.9%.
- 3Y (annualized): 18.4% vs SPY 22.1%; max drawdown -10.8% vs -18.8%; Sharpe 1.11 vs 1.10, effectively a tie.
- 5Y (annualized): 8.1% vs SPY 12.9%; Sharpe 0.39 vs 0.57; max drawdown -26.4% vs -24.5%, worse than SPY.
- Since 2007 (annualized): 7.6% vs SPY 10.9%; Sharpe 0.53 vs 0.55.



Risk and Return by Period: Strategy vs SPY

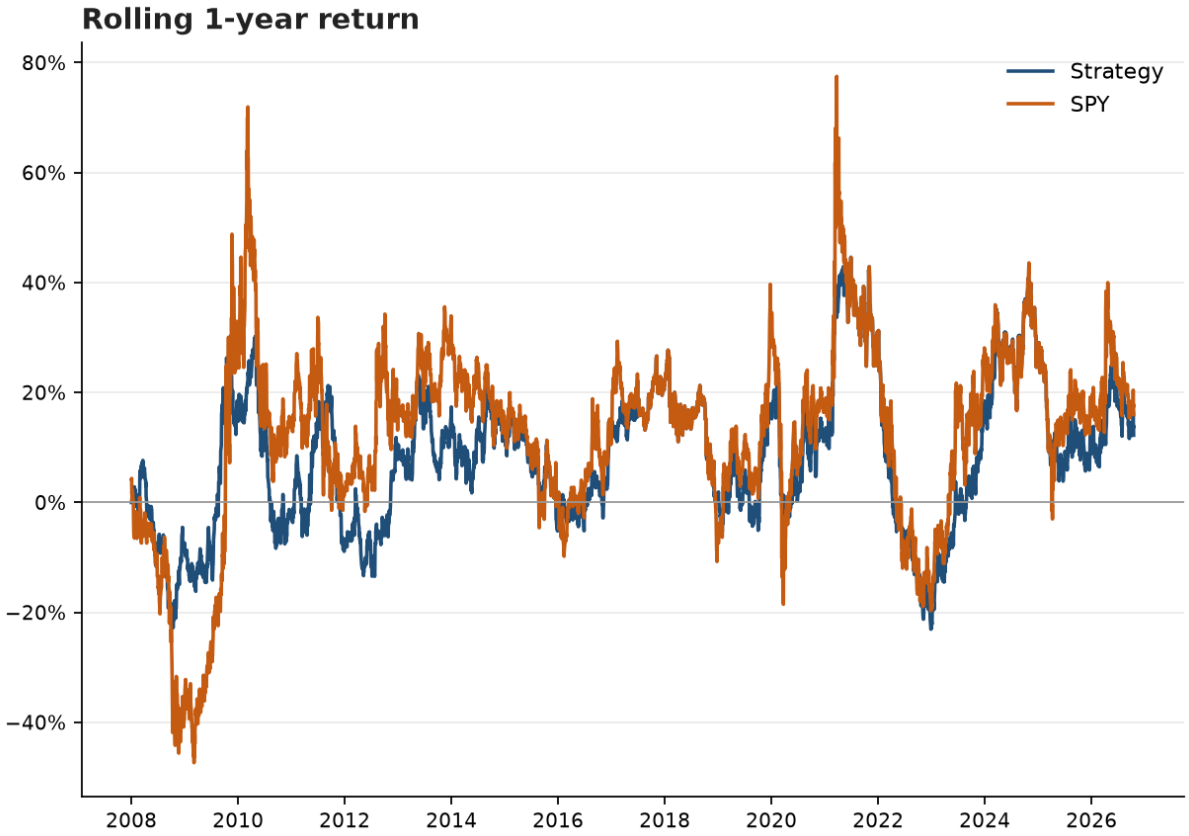
Metric	3M Strat.	3M SPY	1Y Strat.	1Y SPY	3Y Strat.	3Y SPY	5Y Strat.	5Y SPY	SI Strat.	SI SPY
Return (3M total, else annualized)	**1.9%**	**1.9%**	**13.0%**	**16.8%**	**18.4%**	**22.1%**	**8.1%**	**12.9%**	**7.6%**	**10.9%**
Total return (cumulative)	1.9%	1.9%	13.0%	16.8%	65.6%	81.5%	47.6%	83.0%	324.9%	672.5%
Volatility (ann.)	11.1%	11.1%	11.9%	12.7%	12.0%	15.3%	12.3%	17.1%	12.4%	19.5%
Sharpe ratio	n/a	n/a	0.78	1.00	1.11	1.10	0.39	0.57	0.53	0.55
Sortino ratio	n/a	n/a	1.11	1.47	1.57	1.63	0.54	0.83	0.72	0.77
Max drawdown	-3.1%	-3.1%	-6.8%	-8.9%	-10.8%	-18.8%	-26.4%	-24.5%	-26.4%	-55.2%
Calmar ratio	n/a	n/a	1.90	1.89	1.70	1.18	0.31	0.53	0.29	0.20
Beta vs SPY	1.00	1.00	0.87	1.00	0.58	1.00	0.46	1.00	0.29	1.00
Correlation vs SPY	1.00	1.00	0.94	1.00	0.74	1.00	0.65	1.00	0.45	1.00
Up capture	100.0%	100.0%	91.7%	100.0%	83.3%	100.0%	65.7%	100.0%	54.9%	100.0%
Down capture	100.0%	100.0%	94.6%	100.0%	83.3%	100.0%	65.9%	100.0%	53.5%	100.0%
% days positive	44.6%	44.6%	52.6%	52.6%	56.2%	56.4%	53.6%	53.7%	54.1%	55.0%
Best day	1.8%	1.8%	1.9%	2.9%	2.5%	10.5%	2.5%	10.5%	4.4%	14.5%
Worst day	-1.5%	-1.5%	-2.6%	-2.6%	-3.0%	-5.9%	-3.0%	-5.9%	-5.8%	-10.9%
Switches / % time in SPY	0 / 100.0%		2 / 95.2%		8 / 92.2%		30 / 78.2%		158 / 77.0%	

60/40 benchmark	3M	1Y	3Y	5Y	SI
Return (3M total, else ann.)	0.2%	8.3%	14.7%	7.3%	8.4%
Total return (cumulative)	0.2%	8.3%	50.7%	41.7%	389.0%
Volatility (ann.)	7.8%	8.5%	9.7%	11.0%	11.2%
Sharpe ratio	n/a	0.55	1.02	0.35	0.63
Max drawdown	-2.8%	-6.0%	-10.6%	-21.0%	-31.4%
Calmar ratio	n/a	1.39	1.39	0.35	0.27

Periods end on the last completed session. 3M: 2026-07-17 to 2026-10-16, 1Y: 2025-10-17 to 2026-10-16, 3Y: 2023-10-17 to 2026-10-16, 5Y: 2021-10-18 to 2026-10-16, SI: 2007-01-03 to 2026-10-16. SI = since the walk-forward start. Ratios use the 13-week T-bill as the risk-free rate. Current allocation: 100.0% SPY.

Rolling 1-Year Returns: Narrower Range, Usually Behind SPY

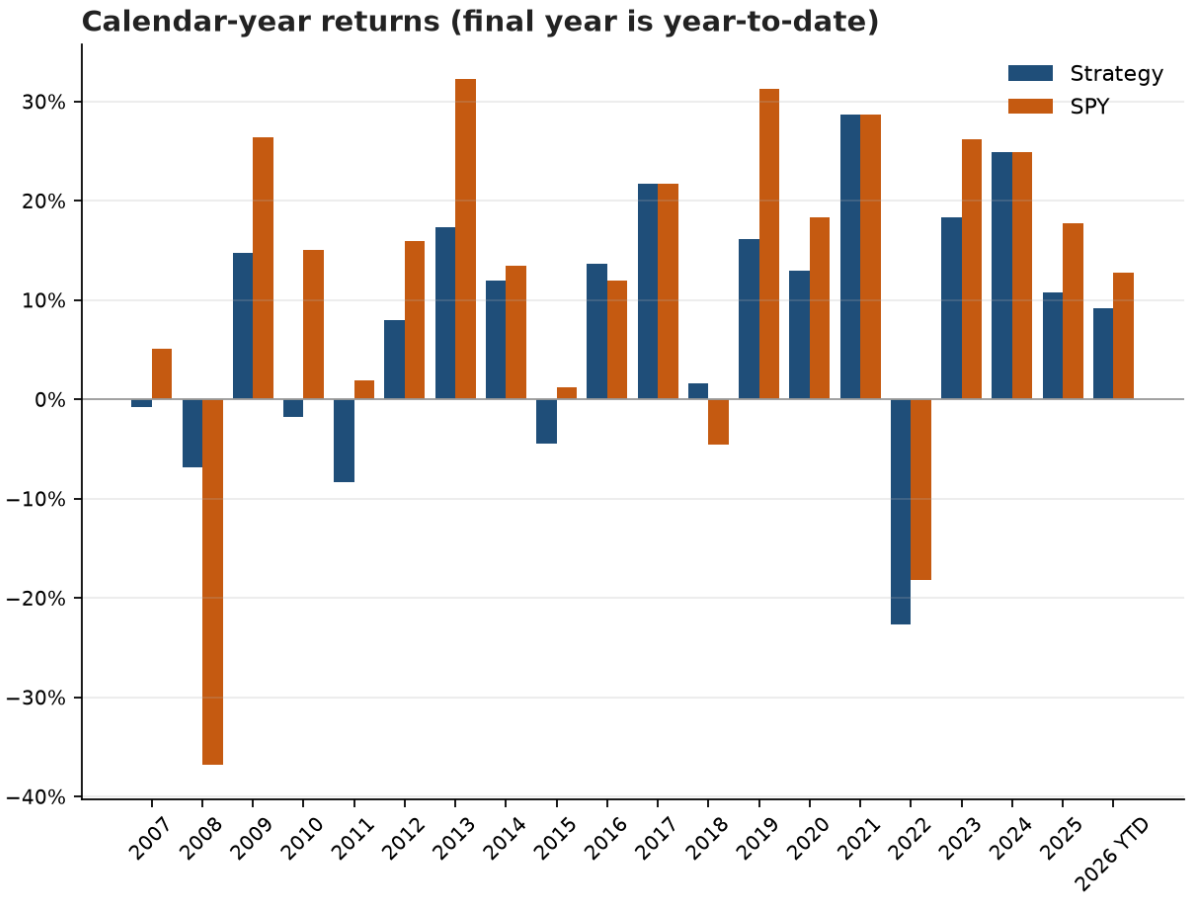
- Across 4728 rolling 1-year periods, the MA strategy beat SPY in 18.7% of them.
- Median rolling 1-year excess return vs SPY: -4.1%.
- Best rolling year 44.6% vs SPY 77.5%; worst rolling year -23.1% vs SPY -47.4%.
- The pattern reflects the trade-off: a narrower range of outcomes in exchange for giving up a meaningful share of SPY's upside.



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When the MA Strategy Lags: Honest Trade-offs

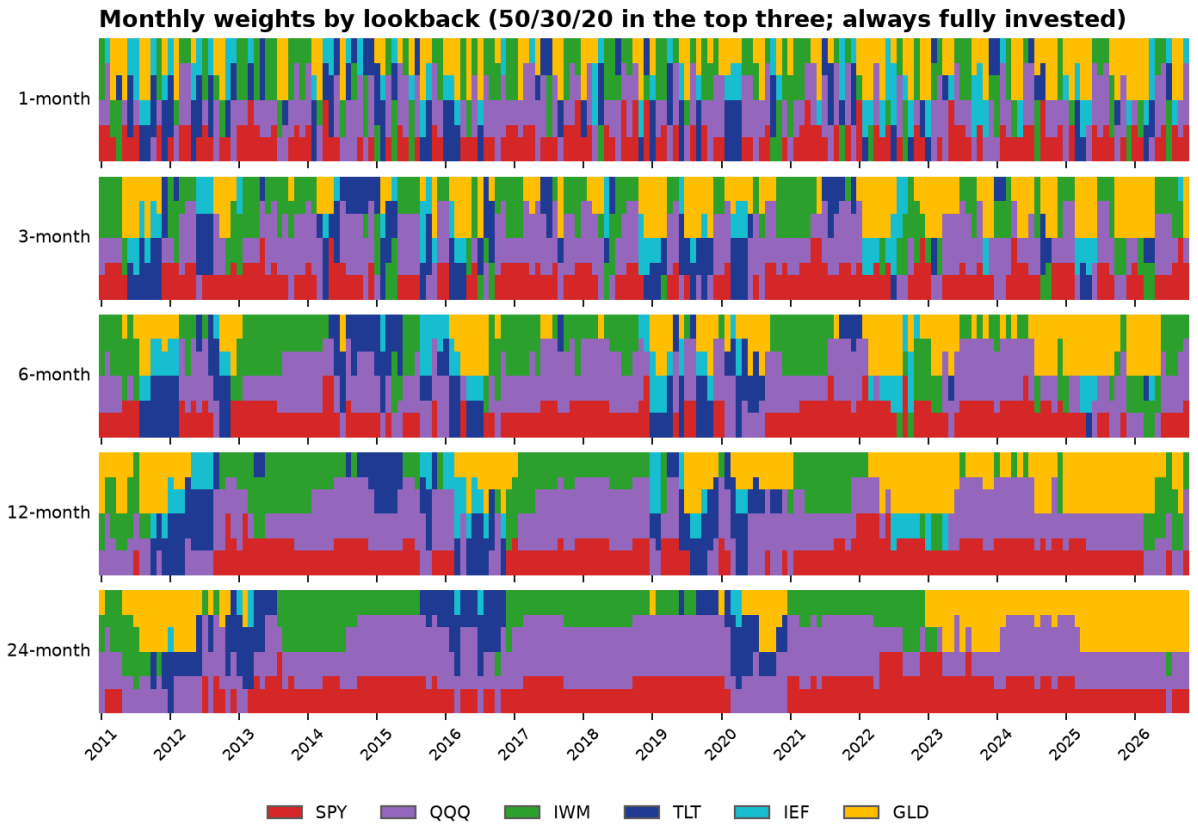
- Since 2007 (SI), the MA strategy was worse than SPY on Sharpe (0.53 vs 0.55) and on CAGR (7.6% vs 10.9%).
- Since 2007 (SI), a 60/40 portfolio beat the MA strategy on CAGR (8.4% vs 7.6%) and Sharpe (0.63 vs 0.53).
- 5Y: worse than SPY on max drawdown (-26.4% vs -24.5%) and Sharpe (0.39 vs 0.57).
- 1Y: worse than SPY on Sharpe (0.78 vs 1.00), with a return of 13.0% vs 16.8%.
- Whipsaw years hurt: 2011 -8.3% vs SPY 1.9%; 2015 -4.5% vs SPY 1.2%; 2022 -22.7% vs SPY -18.2%.
- Strong bull years were often missed in part: 2019 16.2% vs SPY 31.2%; 2025 10.8% vs SPY 17.7%.



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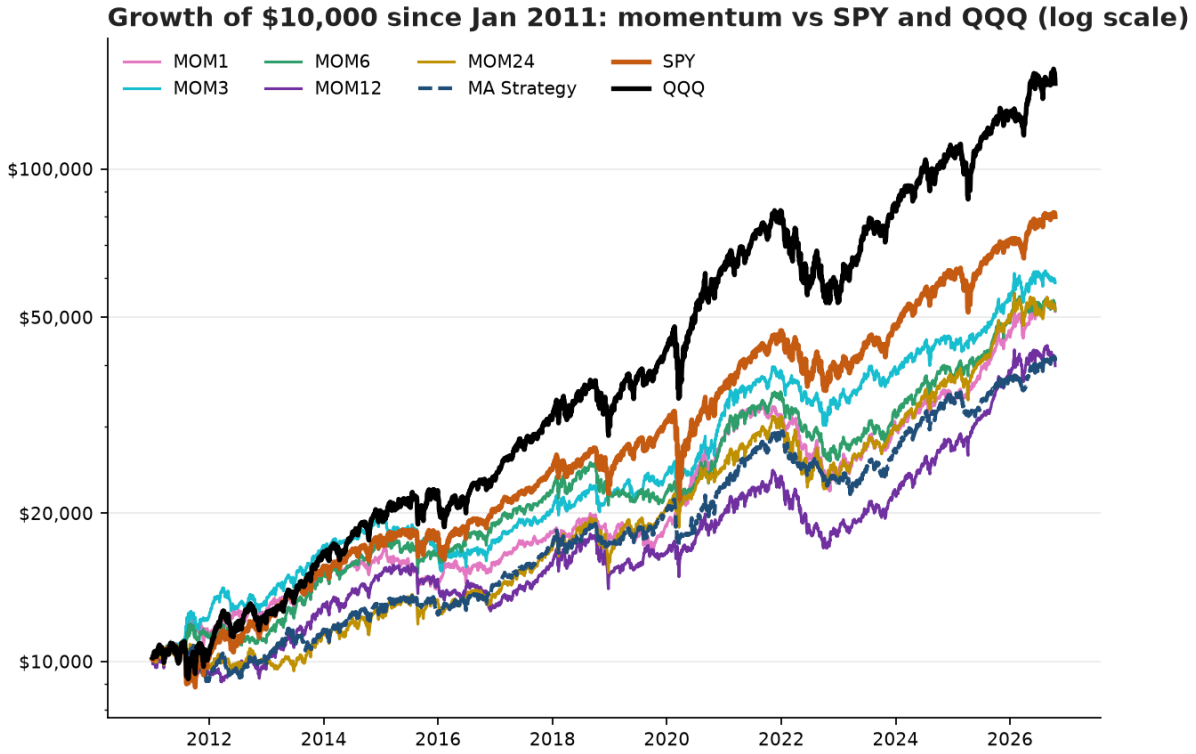
Multi-Asset Momentum: Rules and Monthly Allocations

- Five sub-strategies rank SPY, QQQ, TLT, GLD, IWM and IEF by trailing 1, 3, 6, 12 or 24-month return at each month-end, from January 2011.
- Each holds the top three trailing performers at 50% / 30% / 20%, rebalanced monthly.
- Always fully invested, even when trailing returns are negative; never in cash. There is no defensive exit, and Treasuries are held only if they rank in the top three.
- Average QQQ weight since 2011 ranged from 24.2% (1-Month) to 38.6% (24-Month).
- Results exclude trading costs. Turnover ranged from 21.5x (24-Month) to 106.5x (1-Month) the portfolio.
- Benchmarked against SPY and QQQ buy-and-hold.



Momentum Growth vs SPY and QQQ Since 2011

- Growth of $10,000 since 2011: QQQ $149,098, SPY $79,961, momentum $39,845 (12-Month) to $58,725 (3-Month), before trading costs.
- Every lookback trailed both SPY (14.1%) and QQQ (18.7%) on CAGR since 2011.
- Last 5Y (annualized): momentum 9.4% to 12.1% vs SPY 12.9% and QQQ 14.7%.
- Last 3Y (annualized): momentum 15.9% to 25.1% vs SPY 22.1% and QQQ 25.1%.
- Results varied widely by year: in 2025 the lookbacks returned 24.5% to 41.0% vs SPY 17.7% and QQQ 20.8%.



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1-Month Momentum: Pros and Cons Since 2011

- Pro: Lower volatility than SPY since 2011 (13.7% vs 17.0%), although its Sharpe was lower than SPY's (0.71 vs 0.77).
- Pro: Strong rebound years: 2020 return of 51.2% vs SPY 18.3% and QQQ 48.4%; 2025 return of 35.3% vs SPY 17.7%.
- Con: Worse than SPY since 2011 on CAGR (10.9% vs 14.1%, before costs), and well behind QQQ (18.7%).
- Con: Worse than SPY since 2011 on Sharpe (0.71 vs 0.77); QQQ was higher still at 0.86.
- Con: Always fully invested with no defensive exit; its max drawdown (-32.9%) was effectively a tie with SPY (-33.7%).
- Con: Highest turnover of the lookbacks (106.5x the portfolio); at MA-level costs CAGR falls from 10.9% to 10.6%.

3-Month Momentum: Pros and Cons Since 2011

- Pro: Shallowest observed max drawdown of all lookbacks since 2011 (-24.1% vs SPY -33.7% and QQQ -35.1%), an outcome rather than a defensive rule.
- Pro: 2022 loss of -14.5% vs SPY -18.2% and QQQ -32.6%; 2011 gain of 23.5% vs SPY 1.9%.
- Con: Worse than SPY since 2011 on CAGR (11.9% vs 14.1%, before costs; 11.7% at MA-level costs); QQQ returned 18.7%.
- Con: Worse than SPY since 2011 on Sharpe at full precision; both display as 0.77 (a tie, not an advantage), and QQQ was higher at 0.86.
- Con: Always fully invested with no defensive exit; negative-momentum ETFs can still be held.
- Con: 2015 loss of -12.2% vs SPY 1.2%; turnover of 60.0x the portfolio.

6-Month Momentum: Pros and Cons Since 2011

- Pro: Shallower observed max drawdown than SPY and QQQ since 2011 (-27.8% vs -33.7% and -35.1%).
- Pro: Lower volatility than SPY (13.9% vs 17.0%), although its Sharpe was lower than SPY's (0.71 vs 0.77).
- Con: Worse than SPY since 2011 on CAGR (11.0% vs 14.1%, before costs; 10.8% at MA-level costs); QQQ returned 18.7%.
- Con: Worse than SPY since 2011 on Sharpe (0.71 vs 0.77); QQQ was higher at 0.86.
- Con: Always fully invested with no defensive exit; negative-momentum ETFs can still be held.
- Con: 2022 loss of -25.4% vs SPY -18.2%; turnover of 45.7x the portfolio.

12-Month Momentum: Pros and Cons Since 2011

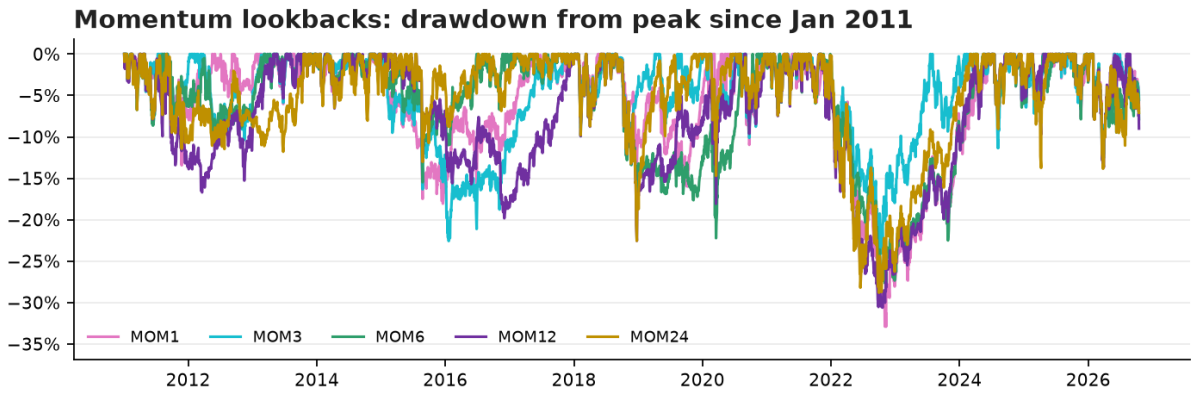
- Pro: Shallower observed max drawdown than SPY and QQQ since 2011 (-30.6% vs -33.7% and -35.1%).
- Pro: 2025 return of 41.0% vs SPY 17.7% and QQQ 20.8%.
- Con: Worse than SPY since 2011 on CAGR (9.2% vs 14.1%, before costs); QQQ returned 18.7%.
- Con: Worse than SPY since 2011 on Sharpe (0.58 vs 0.77), the lowest Sharpe of the lookbacks; QQQ was 0.86.
- Con: Always fully invested with no defensive exit; negative-momentum ETFs can still be held.
- Con: At MA-level costs its CAGR of 9.1% is effectively tied with the MA strategy (9.3%); 2016 return -10.7% vs SPY 12.0%.

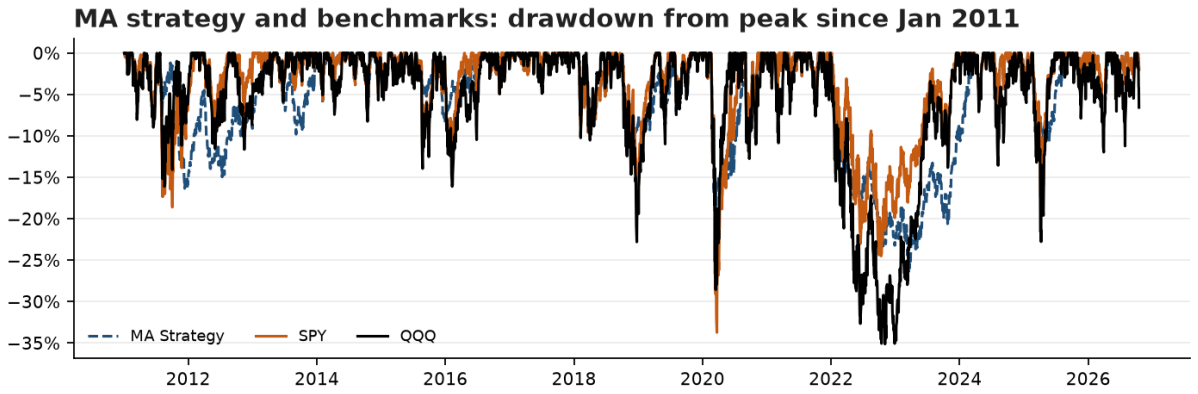
24-Month Momentum: Pros and Cons Since 2011

- Pro: Shallower observed max drawdown than SPY and QQQ since 2011 (-28.8% vs -33.7% and -35.1%).
- Pro: Lowest turnover of the lookbacks (21.5x the portfolio); at MA-level costs CAGR is 10.9% vs 11.0% before costs.
- Con: Worse than SPY since 2011 on CAGR (11.0% vs 14.1%, before costs); QQQ returned 18.7%.
- Con: Worse than SPY since 2011 on Sharpe (0.66 vs 0.77); QQQ was 0.86.
- Con: Always fully invested with no defensive exit; negative-momentum ETFs can still be held.
- Con: Most equity-like lookback: correlation 0.74 and down capture 80.8% vs SPY; 2022 return -24.0% vs SPY -18.2%.

Momentum Drawdowns: Observed Outcomes vs SPY and QQQ

- Observed max drawdowns since 2011: -24.1% (3-Month) to -32.9% (1-Month) vs SPY -33.7% and QQQ -35.1%. These are outcomes, not the result of a defensive rule.
- In 2022, the lookbacks returned -14.5% to -25.4% vs SPY -18.2% and QQQ -32.6%; four of the five lost more than SPY.
- Down capture vs SPY ranged from 52.3% (1-Month) to 80.8% (24-Month).
- Momentum never moves to cash; drawdown depth depends on how the three top-ranked ETFs behave.

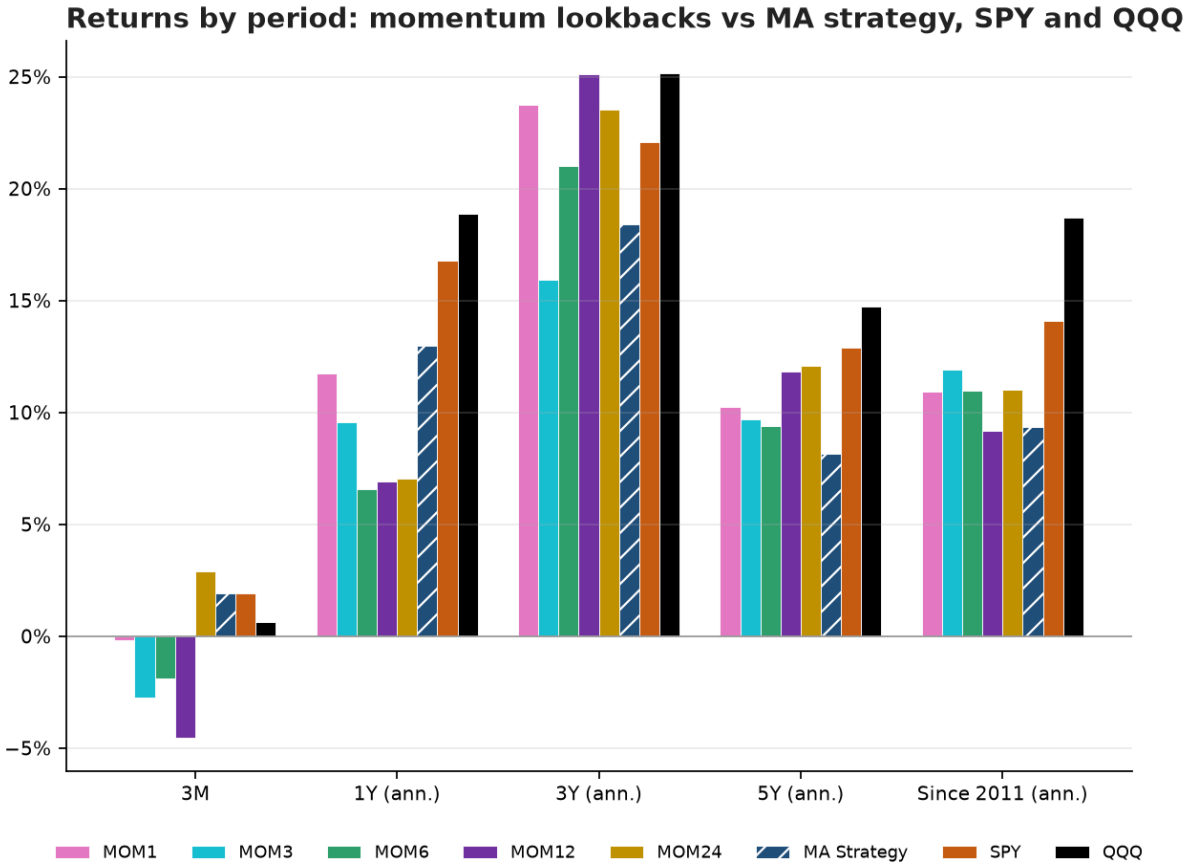




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Momentum vs MA Strategy vs SPY and QQQ Since 2011

- CAGR since 2011: QQQ 18.7%, SPY 14.1%, momentum 9.2% to 11.9% before trading costs, MA strategy 9.3% net of costs.
- Like-for-like at MA-level costs, momentum CAGR was 9.1% to 11.7% vs MA 9.3%; 12-Month (9.1%) ties the MA strategy, the other four lookbacks were higher.
- Sharpe: QQQ 0.86, SPY 0.77, momentum 0.58 to 0.77 (before costs), MA 0.67 (net). No strategy beat SPY's Sharpe; QQQ was highest.
- Max drawdown: 3-Month momentum -24.1% (before costs) and MA -26.4% vs SPY -33.7% and QQQ -35.1%; 1-Month (-32.9%) roughly matched SPY.
- Volatility: MA strategy 12.1%, momentum 13.7% to 15.2%, SPY 17.0%, QQQ 20.7%.
- Momentum excludes trading costs; the MA strategy is net of modeled costs. Only the MA strategy has a defensive rule.



Head to Head Since 2011: Momentum, MA, SPY and QQQ

Return (3M total, else annualized)	3M	1Y	3Y	5Y	Since 2011
1-month momentum	-0.2%	11.7%	23.7%	10.3%	10.9%
3-month momentum	-2.8%	9.5%	15.9%	9.7%	11.9%
6-month momentum	-1.9%	6.6%	21.0%	9.4%	11.0%
12-month momentum	-4.5%	6.9%	25.1%	11.8%	9.2%
24-month momentum	2.9%	7.1%	23.5%	12.1%	11.0%
MA trend strategy	1.9%	13.0%	18.4%	8.1%	9.3%
SPY (buy and hold)	1.9%	16.8%	22.1%	12.9%	14.1%
QQQ (buy and hold)	0.7%	18.9%	25.1%	14.7%	18.7%

Since 2011	CAGR	Total	Vol.	Sharpe	Sortino	Max DD	Calmar	Beta	Corr.	Up cap.	Down cap.
1-month momentum	10.9%	413.7%	13.7%	0.71	1.00	-32.9%	0.33	0.37	0.45	56.0%	52.3%
3-month momentum	11.9%	487.3%	13.9%	0.77	1.07	-24.1%	0.49	0.38	0.47	60.2%	56.1%
6-month momentum	11.0%	416.2%	13.9%	0.71	0.99	-27.8%	0.40	0.48	0.59	63.5%	60.9%
12-month momentum	9.2%	298.5%	14.3%	0.58	0.80	-30.6%	0.30	0.55	0.65	67.5%	67.6%
24-month momentum	11.0%	418.0%	15.2%	0.66	0.92	-28.8%	0.38	0.67	0.74	80.6%	80.8%
MA trend strategy	9.3%	307.5%	12.1%	0.67	0.91	-26.4%	0.35	0.41	0.58	65.3%	65.1%
SPY (buy and hold)	14.1%	699.6%	17.0%	0.77	1.08	-33.7%	0.42	1.00	1.00	100.0%	100.0%
QQQ (buy and hold)	18.7%	1391.0%	20.7%	0.86	1.22	-35.1%	0.53	1.13	0.93	118.1%	115.5%

Common period 2011-01-03 to 2026-10-16. Momentum: hold 50% / 30% / 20% in the top three trailing performers of SPY, QQQ, TLT, GLD, IWM, IEF each month, rebalanced monthly; always fully invested, even when trailing returns are negative (never in cash). **Momentum excludes trading costs; the MA strategy is net of modeled costs.** Ratios use the 13-week T-bill as the risk-free rate. Beta, correlation and capture are measured vs SPY.

Momentum Detail: Calendar-Year Returns and Growth of $10,000

Year	1-month	3-month	6-month	12-month	24-month	MA	SPY	QQQ
2011	14.9%	23.5%	12.9%	-5.1%	-2.6%	-8.3%	1.9%	3.5%
2012	8.3%	7.2%	-2.0%	3.6%	2.3%	8.0%	16.0%	18.1%
2013	24.4%	30.2%	36.7%	33.2%	15.3%	17.4%	32.3%	36.6%
2014	4.6%	12.1%	14.0%	14.0%	10.5%	12.0%	13.5%	19.2%
2015	-12.0%	-12.2%	-4.6%	-3.5%	3.9%	-4.5%	1.2%	9.4%
2016	9.9%	2.5%	14.4%	-10.7%	2.2%	13.6%	12.0%	7.1%
2017	16.3%	15.9%	17.1%	21.7%	24.9%	21.7%	21.7%	32.7%
2018	-1.7%	9.0%	-1.3%	-4.0%	-3.5%	1.6%	-4.6%	-0.1%
2019	5.2%	9.0%	1.1%	10.3%	25.6%	16.2%	31.2%	39.0%
2020	51.2%	33.5%	29.2%	21.0%	22.5%	12.9%	18.3%	48.4%
2021	10.8%	21.0%	23.2%	17.4%	24.5%	28.7%	28.7%	27.4%
2022	-23.9%	-14.5%	-25.4%	-22.7%	-24.0%	-22.7%	-18.2%	-32.6%
2023	26.5%	19.5%	23.4%	22.1%	27.1%	18.3%	26.2%	54.9%
2024	10.7%	9.9%	16.7%	20.4%	23.1%	24.9%	24.9%	25.6%
2025	35.3%	24.5%	29.4%	41.0%	32.7%	10.8%	17.7%	20.8%
2026 YTD	12.9%	8.7%	6.1%	5.5%	5.7%	9.1%	12.8%	15.9%

Growth of $10,000 since 2011	Value
1-month momentum	$51,370
3-month momentum	$58,725
6-month momentum	$51,623
12-month momentum	$39,845
24-month momentum	$51,804
MA trend strategy	$40,753
SPY (buy and hold)	$79,961
QQQ (buy and hold)	$149,098

Calendar-year returns are total returns; the final year is year-to-date through 2026-10-16. Momentum excludes trading costs; MA = moving-average trend strategy, net of modeled costs.

Momentum Detail: Allocations, Activity and Cost Sensitivity

Average weight since 2011	SPY	QQQ	TLT	GLD	IWM	IEF
1-month momentum	16.7%	24.2%	13.4%	17.2%	18.6%	9.9%
3-month momentum	18.1%	26.8%	12.1%	17.7%	18.9%	6.4%
6-month momentum	18.9%	28.6%	11.1%	16.1%	19.1%	6.3%
12-month momentum	18.7%	31.4%	10.2%	18.1%	16.7%	4.9%
24-month momentum	22.2%	38.6%	8.8%	14.7%	14.7%	0.8%

Current weights (2026-10)	SPY	QQQ	TLT	GLD	IWM	IEF
1-month momentum	30.0%	50.0%	-	-	-	20.0%
3-month momentum	30.0%	20.0%	-	50.0%	-	-
6-month momentum	30.0%	50.0%	-	-	20.0%	-
12-month momentum	20.0%	50.0%	-	-	30.0%	-
24-month momentum	20.0%	30.0%	-	50.0%	-	-

Activity since 2011	Months	Rebalances	Turnover*	CAGR as shown	CAGR at 5 bps
1-month momentum	190	181	106.5x	10.9%	10.6%
3-month momentum	190	154	60.0x	11.9%	11.7%
6-month momentum	190	139	45.7x	11.0%	10.8%
12-month momentum	190	117	33.6x	9.2%	9.1%
24-month momentum	190	88	21.5x	11.0%	10.9%

*Turnover = how many times the whole portfolio was traded since 2011 (e.g. 50.0x) (sum of monthly target-weight changes, halved; the initial purchase counts as 1.0). "CAGR at 5 bps" charges that cost per unit of turnover; momentum results elsewhere exclude trading costs. Weights are monthly targets; holdings drift within each month.

Portfolio Fit: Who Each Strategy May Suit

- MA strategy: for investors who want SPY exposure with a rules-based Treasury exit and accept lower long-run return; beta 0.29 and correlation 0.45 to SPY since 2007.
- Momentum: a fully invested multi-asset rotation across equities, Treasuries and gold; beta 0.37 to 0.67 to SPY since 2011, with no defensive exit.
- Neither suits investors seeking full equity upside: MA up capture was 54.9% since 2007, and every momentum lookback trailed SPY and QQQ on CAGR.
- Best used as a satellite alongside core equity holdings, sized with the understanding that SPY led on return since 2007 and since 2011.
- Momentum figures are before trading costs; implementation costs and taxes should be assessed for each mandate.

Hypothetical backtested performance. Not actual trading; not net of fees. MA strategy results are net of modeled trading costs; momentum results exclude trading costs. Past performance does not guarantee future results. See Important Disclosures.

Next Steps

- Request the full methodology document, including the trading-cost model and walk-forward selection code.
- Request daily return series for independent verification of the backtests.
- Discuss implementation details: execution, rebalancing timing and tax considerations for your mandate.
- Arrange a meeting with the investment team to discuss sizing within your portfolio.

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Meet the Digital Investment Team (AI-generated, fictional)

Margaret Okonkwo-Hale

Lead Portfolio Manager

Focus: Overall strategy oversight, allocation decisions and investment committee leadership

Certifications: CFA, CAIA | **Experience:** 26 yrs

Education

BA, Economics, University of Michigan (1999)
MBA, Finance, University of Chicago Booth School of Business (2005)

Experience

Analyst, Global Asset Allocation, Harborview Capital Partners (2000-2005)
Senior Portfolio Manager, Ridgeline Tactical Advisors (2005-2015)
Head of Multi-Asset Strategies, Northgate Systematic Investments (2015-2026)

Margaret Okonkwo-Hale has spent more than two decades building and running rules-based allocation portfolios across equities, Treasuries and real assets. On this team she owns final responsibility for both the SPY/IEF trend rotation and the multi-asset momentum strategy, chairing the investment committee and approving any change to rules or universe. Her distinctive strength is discipline: she insists that every model adjustment be documented, tested out of sample and justified on economic grounds before it reaches live portfolios, which keeps the process transparent for investors.

AI-generated fictional persona

Daniel Reyes-Lindqvist

Senior Quantitative Research Analyst

Focus: Quantitative research and signal design for moving-average and momentum rules

Certifications: CQF | **Experience:** 13 yrs

Education

BS, Mathematics, University of Texas at Austin (2008)
PhD, Statistics, Carnegie Mellon University (2013)

Experience

Quantitative Researcher, Bluefin Quantitative Research (2013-2019)
Senior Quant Analyst, Northgate Systematic Investments (2019-2026)

Daniel Reyes-Lindqvist came to investing from academic statistics, where he studied time-series estimation under noisy data. He owns signal design for both strategies, including moving-average lookback selection for the SPY/IEF rotation and ranking methodology for the monthly momentum universe. His distinctive strength is skepticism toward overfitting: he routinely stress-tests parameter choices across neighbouring settings and subperiods, favouring robust, simple rules over finely tuned ones that look impressive only in hindsight.

AI-generated fictional persona

Priya Natarajan

Risk Management Analyst

Focus: Portfolio risk, drawdown monitoring and stress testing

Certifications: FRM, PRM | **Experience:** 13 yrs

Education

BCom, Finance, University of Toronto (2011)
MS, Financial Engineering, Columbia University (2013)

Experience

Risk Analyst, Cedarbrook Asset Management (2013-2018)
Senior Risk Associate, Silverline Risk Partners (2018-2022)
Risk Management Analyst, Northgate Systematic Investments (2022-2026)

Priya Natarajan trained in financial engineering and has worked in independent risk functions throughout her career. She owns risk oversight for the team, monitoring volatility, concentration and drawdown behaviour across both strategies and running scenario analyses for events such as simultaneous equity and bond selloffs. Her distinctive strength is translating abstract risk metrics into plain-language scenarios, helping the committee understand how the strategies might behave when trend signals whipsaw or correlations break down.

AI-generated fictional persona

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Digital Investment Team, continued (AI-generated, fictional)

Thomas Achterberg

Macro and Rates Analyst

Focus: Macro and interest rates, Treasury duration exposure (IEF and TLT)

Certifications: CFA | **Experience:** 19 yrs

Education

BA, Economics, Georgetown University (2004)
MA, International Economics, Johns Hopkins University SAIS (2007)

Experience

Rates Strategist, Meridian Fixed Income Advisors (2007-2014)
Senior Macro Analyst, Eastwater Capital Management (2014-2021)
Macro and Rates Analyst, Northgate Systematic Investments (2021-2026)

Thomas Achterberg spent his early career as a Treasury rates strategist before moving into macro research for multi-asset portfolios. He owns the fixed income side of the team's work, analysing how IEF and TLT exposures respond to inflation, policy shifts and changes in the stock-bond correlation. His distinctive strength is providing economic context for systematic signals: he explains why defensive bond allocations may help or hurt in different rate regimes without overriding the rules themselves.

AI-generated fictional persona

Kenji Watanabe-Cole

Market Data and Backtest Engineer

Focus: Market data engineering and backtest integrity

Certifications: None | **Experience:** 9 yrs

Education

BS, Computer Science, University of Washington (2017)

Experience

Data Engineer, Lattice Data Systems (2017-2021)
Market Data and Backtest Engineer, Northgate Systematic Investments (2021-2026)

Kenji Watanabe-Cole is a software engineer who built financial data pipelines before joining the investment side. He owns the team's price and total-return data, dividend adjustments, and the backtesting framework used for both strategies. His distinctive strength is backtest hygiene: he guards rigorously against look-ahead bias, survivorship issues and misaligned rebalance dates, and maintains reproducible, version-controlled research so any historical result can be regenerated and audited by colleagues or investors.

AI-generated fictional persona

Elena Vasquez-Moreau

Trading and Investor Communications Analyst

Focus: Trading, execution and investor communications

Certifications: CMT | **Experience:** 7 yrs

Education

BA, Communications and Economics, Boston College (2019)

Experience

Trading Associate, Fairhaven Brokerage Services (2019-2022)
Trading and Communications Analyst, Northgate Systematic Investments (2022-2026)

Elena Vasquez-Moreau began her career on an ETF trading desk, learning execution mechanics, spreads and closing-auction dynamics. She owns implementation of monthly rebalances and rotation trades, minimising slippage across SPY, QQQ, IWM, GLD, TLT and IEF, and also drafts investor letters explaining position changes. Her distinctive strength is bridging operations and communication: she can describe exactly what was traded and why in clear, jargon-free language that investors find easy to follow.

AI-generated fictional persona

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Supporting Data: Calendar Years and Rolling Returns

Year	Strategy	SPY	60/40
2007	-0.7%	5.1%	7.7%
2008	-6.8%	-36.8%	-16.7%
2009	14.7%	26.4%	13.3%
2010	-1.8%	15.1%	13.4%
2011	-8.3%	1.9%	8.3%
2012	8.0%	16.0%	11.3%
2013	17.4%	32.3%	15.6%
2014	12.0%	13.5%	12.0%
2015	-4.5%	1.2%	1.8%
2016	13.6%	12.0%	7.8%
2017	21.7%	21.7%	13.8%
2018	1.6%	-4.6%	-2.0%
2019	16.2%	31.2%	21.8%
2020	12.9%	18.3%	16.9%
2021	28.7%	28.7%	15.1%
2022	-22.7%	-18.2%	-16.4%
2023	18.3%	26.2%	17.0%
2024	24.9%	24.9%	14.2%
2025	10.8%	17.7%	14.3%
2026 YTD	9.1%	12.8%	6.3%

Rolling 1-year returns	Strategy	SPY
Best 1-year period	44.6%	77.5%
Worst 1-year period	-23.1%	-47.4%
Periods beating SPY	18.7%	
Median excess vs SPY	-4.1%	
Growth of $10,000 (since walk-forward start)		
Strategy	$42,495	
SPY	$77,253	
60/40	$48,905	

Calendar-year returns are total returns. The final year is year-to-date through 2026-10-16. Rolling figures cover every 252-trading-day window since the walk-forward start. 60/40 is rebalanced daily, without costs.

Important Disclosures

Hypothetical performance. All performance shown is hypothetical and backtested. It does not represent actual trading or actual client accounts. Backtested results are prepared with the benefit of hindsight and have inherent limitations. No representation is made that any account will or is likely to achieve similar results.

Methodology. The strategy holds either SPY or IEF, based on whether SPY closed above a moving average the previous day. The moving average (5, 20, 50 or 200 days) is chosen each January using only data available before that date (walk-forward). Results include a modeled trading cost of 5 basis points per full switch between SPY and IEF. They exclude management fees, taxes and market impact beyond that assumption, and use adjusted closing prices, which include dividends. Partial calendar years are year-to-date.

Momentum methodology. Five monthly momentum sub-strategies (1, 3, 6, 12 and 24-month lookbacks) rank the six ETFs by trailing return at each month-end and, from January 2011, hold 50% / 30% / 20% in the top three trailing performers of SPY, QQQ, TLT, GLD, IWM, IEF each month, rebalanced monthly; always fully invested, even when trailing returns are negative (never in cash). QQQ and SPY buy-and-hold are shown as benchmarks. Momentum results **do not include trading costs**, while the moving-average strategy is shown net of modeled costs, so comparisons between them favor momentum by the cost of its switches.

In-sample results. Comparisons of individual moving-average rules over the full period are in-sample. They are shown for context only, and overstate what an investor could have known in advance.

Risk. Investing involves risk, including loss of principal. Trend-following signals can lag reversals, cause frequent switching in choppy markets, and underperform buy-and-hold for long periods. Treasury ETFs carry interest-rate risk and can fall alongside equities.

Data. Prices from Yahoo Finance via the yfinance library. The data has not been independently verified. The risk-free rate is the 13-week Treasury bill yield (^IRX).

Investment team. The investment team members in this presentation are AI-generated fictional personas created by an AI system for illustration. They are not real people, and their stated employers are fictional.

Not advice. This presentation was produced by an automated multi-agent research prototype. It is for informational purposes only, and is not investment advice, an offer, or a solicitation to buy or sell any security.